

Lumora

AI-Powered Medical Image Analysis Platform

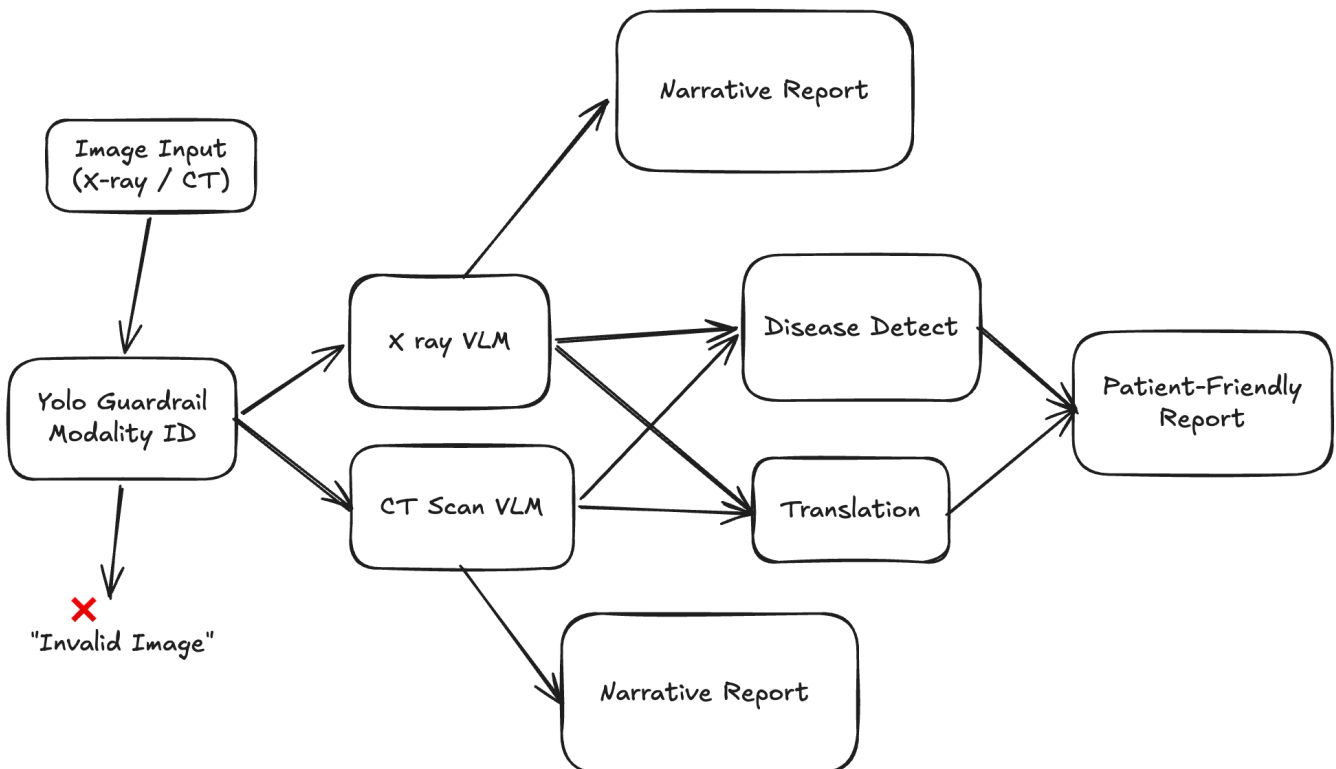
A clinical decision-support tool that analyzes chest X-rays and CT scans using Vision-Language Models (VLMs). Lumora generates narrative radiology reports, detects 26 pathologies with a multi-label classifier, and translates complex clinical findings into patient-friendly language — all behind an intelligent image modality guardrail.

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Architecture Overview

Lumora processes medical images through a sequential four-stage pipeline, where each stage is powered by a purpose-built deep learning model:



Stage	Model	Purpose
1. Guardrail	YOLOv11s-cls	Validates that the uploaded image is a legitimate chest X-ray or CT scan before consuming compute
2. Report Generation	DenseNet-121 + GPT-2 VLM	Encodes the medical image into visual embeddings and autoregressively generates a narrative radiology report

Stage	Model	Purpose
3. Disease Detection	ClinicalBERT	Performs multi-label classification on the generated report, extracting 26 pathology labels
4. Translation	T5-small + LoRA	Rewrites the clinical jargon into clear, patient-friendly language

Model Specifications

Complete specification matrix for all five models in the Lumora inference pipeline.

Specification	1. Modality Identification Model (YOLO)	2. X-Ray VLM	3. CT VLM	4. Translation Model (T5)	5. Disease Detection Model (ClinicalBERT)
Primary Function	Image input validation / guardrail	Frontal chest X-ray narrative report generation	Converted 2D CT slice narrative report generation	Clinical jargon translation to patient friendly layperson terms	Multi-label extraction of 26 pathologies from radiology reports
Transfer Learning Base	yolo11s-cls.pt (Ultralytics)	densenet121 + gpt2	densenet121 + gpt2	t5-small + LoRA adapter	emilyalsentzer/Bio_ClinicalBERT
Parameter Count	~2.8 Million	~133 Million	~133 Million	~60.8 Million (LoRA: 295k)	~110 Million
Activation Function	SiLU	GELU (Decoder), ReLU (Encoder)	GELU (Decoder), ReLU (Encoder)	ReLU	GELU
Dropout Rate	0.0	0.1 (Decoder), 0.0 (Encoder)	0.1 (Decoder), 0.0 (Encoder)	0.1 (LoRA & base T5)	0.1 (hidden & attention)
Loss Function	CrossEntropyLoss	CrossEntropyLoss (shifted logits)	CrossEntropyLoss (shifted logits)	CrossEntropyLoss (Seq2Seq target)	Binary Cross Entropy with Logits (BCEWithLogits Loss)
Optimizer	Auto-selected (SGD/AdamW)	AdamW	AdamW	AdamW	AdamW
Learning Rate	0.01 (initial lr0)	Phase 1: 1e-4, Phase 2: 2e-5	Phase 1: 1e-4, Phase 2: 2e-5	3e-4	3e-5 (with 10% warmup, cosine decay)
Weight Decay	0.0005	Default (0.01)	Default (0.01)	Default (0.01)	0.01
Accumulator Steps	1	1	1	4 (effective batch size: 16)	1
Batch Size	128	8 (run) / 4 (local script)	4	4	8

Specification	1. Modality Identification Model (YOLO)	2. X-Ray VLM	3. CT VLM	4. Translation Model (T5)	5. Disease Detection Model (ClinicalBERT)
Epochs	100	Phase 1: 3, Phase 2: 3 (Total: 6)	Phase 1: 3, Phase 2: 3 (Total: 6)	5	12
Input Shape / Size	224 × 224 (image)	224 × 224 (image), max 128 tokens	224 × 224 (image), max 128 tokens	max input 512, target 256 tokens	max 512 tokens
Training Dataset Size	2,640 images (80%)	64,592 samples	2,301 samples	840 samples	840 samples
Validation /Test Size	660 images (20%)	504 samples (Val)	106 samples (Val)	105 (Val) / 105 (Test)	105 (Val) / 105 (Test)
Hugging Face Hub URL	pranto24/xray_ct_scan_identification_model	nur9211/mimic-vlm-model	nur9211/ct-rate-vlm-model	nur9211/lumora_translation	nur9211/lumora_disease_classifier

Individual Model Details

1. Modality Identification Model (YOLO Guardrail)

The first line of defense in the pipeline. A **YOLOv11s-cls** image classifier trained on 3,300 images across three categories: `xray`, `ct_scan`, and `other`. It prevents non-medical images (selfies, documents, etc.) from reaching the computationally expensive VLM stage.

- **Architecture:** Ultralytics YOLOv11 small classification backbone with SiLU activations
- **Pre-screen:** An additional variance-based heuristic rejects blank or near-uniform images before the YOLO forward pass
- **Inference output:** Class label + confidence score; only `xray` or `ct_scan` labels above threshold proceed

2. X-Ray Vision-Language Model

The core report generation model for **frontal chest X-rays**. Built as a custom multimodal architecture combining a DenseNet-121 visual encoder with a GPT-2 language decoder, bridged by a learned linear projector.

- **Encoder:** `DenseNet-121` (pretrained on ImageNet) extracts 1,024-dim visual features from 224×224 images
- **Projector:** A linear layer maps the 1,024-dim feature vector to the 768-dim GPT-2 embedding space
- **Decoder:** `GPT-2` autoregressively generates radiology reports conditioned on the projected visual embedding
- **Training:** Two-phase strategy — Phase 1 freezes the encoder and trains only the projector + decoder; Phase 2 fine-tunes the entire network end-to-end at a lower learning rate
- **Dataset:** Trained on **MIMIC-CXR** (64,592 image-report pairs from the Medical Information Mart for Intensive Care)

3. CT Vision-Language Model

An architecturally identical VLM to the X-ray model, separately fine-tuned for **CT scan** report generation.

- **Input adaptation:** 3D NIfTI CT volumes are converted to representative 2D slices during preprocessing
- **Architecture:** Same `DenseNet-121` → `Projector` → `GPT-2` pipeline as the X-ray VLM
- **Training:** Same two-phase regime; trained on **CT-RATE** dataset (2,301 samples)
- **Use case:** Generates narrative reports for axial CT slice imagery covering thoracic anatomy

4. Translation Model (T5 + LoRA)

Rewrites clinical radiology reports into language that patients and non-specialists can understand.

- **Base model:** `T5-small` (60.8M parameters) — a sequence-to-sequence transformer
- **Fine-tuning:** Parameter-efficient fine-tuning with **LoRA** (Low-Rank Adaptation) — only 295K trainable parameters
- **Prompt format:** Input is prefixed with `"translate to patient-friendly: "` before the clinical text
- **Decoding:** Beam search with `num_beams=4` for fluent, coherent output
- **Dataset:** 840 curated clinical → patient-friendly report pairs

5. Disease Detection Model (ClinicalBERT)

A multi-label text classifier that extracts structured pathology labels from the generated narrative report.

- **Base model:** `Bio_ClinicalBERT` — a BERT variant pre-trained on clinical notes from MIMIC-III
- **Classification head:** A linear layer mapping the `[CLS]` token representation to 26 pathology logits
- **Output:** Sigmoid activation produces independent probabilities for each of the 26 disease labels
- **Threshold:** A recall-optimized threshold of `0.15` is applied at inference time
- **Post-processing:** Mutual-exclusion logic resolves contradictions (e.g., pathologies present ↔ "No acute disease")

26 Supported Pathology Labels — Full Dataset Breakdown

The ClinicalBERT classifier is trained on reports from **both** the MIMIC-CXR (X-Ray) and CT-RATE (CT) datasets combined. The 26 canonical labels were selected as those appearing ≥ 8 times across the unified training corpus.

X-Ray (MIMIC-CXR) — Primary Disease Coverage

Diseases commonly observed in frontal chest radiograph reports from the MIMIC-CXR dataset:

#	Disease Label	Frequency (approx.)
1	Atelectasis	Very High
2	Cardiomegaly	Very High
3	Pleural Effusion	Very High

#	Disease Label	Frequency (approx.)
4	Pneumonia	High
5	Pneumothorax	High
6	Pulmonary Edema/Vascular Congestion	High
7	Consolidation	Moderate
8	No Acute Cardiopulmonary Disease	Very High
9	Pulmonary Fibrosis/Scarring	Moderate
10	Pulmonary Nodules	Moderate
11	Rib/Bone Fracture	Moderate
12	Possible Aspiration	Low–Moderate

CT Scan (CT-RATE) — Extended Disease Coverage

Diseases additionally observed in thoracic CT reports from the CT-RATE dataset (CT captures deeper anatomical structures beyond chest X-rays):

#	Disease Label	Frequency (approx.)
1	Atherosclerosis	High
2	Emphysema/COPD	High
3	Hepatic Steatosis	Moderate
4	Hiatal Hernia	Moderate
5	Pericardial Effusion	Moderate
6	Mosaic Attenuation Pattern	Moderate
7	Aortic Dilation	Low–Moderate
8	Lymphadenopathy	Low–Moderate
9	Cholelithiasis	Low–Moderate
10	Osteoporosis	Low–Moderate
11	Spinal Degenerative Changes	Moderate
12	Mild Scoliosis	Low–Moderate
13	Pulmonary Artery Enlargement	Low–Moderate
14	Possible Malignancy/Mass	Low

Combined 26-Label Vocabulary (ClinicalBERT Output Classes)

All 26 labels used as multi-label output classes during ClinicalBERT training, with dataset origin noted:

#	Pathology	Primary Dataset Source
1	Aortic Dilation	CT-RATE

#	Pathology	Primary Dataset Source
2	Atelectasis	MIMIC-CXR + CT-RATE
3	Atherosclerosis	CT-RATE
4	Cardiomegaly	MIMIC-CXR + CT-RATE
5	Cholelithiasis	CT-RATE
6	Consolidation	MIMIC-CXR + CT-RATE
7	Emphysema/COPD	CT-RATE
8	Hepatic Steatosis	CT-RATE
9	Hiatal Hernia	CT-RATE
10	Lymphadenopathy	CT-RATE
11	Mild Scoliosis	CT-RATE
12	Mosaic Attenuation Pattern	CT-RATE
13	No Acute Cardiopulmonary Disease	MIMIC-CXR + CT-RATE
14	Osteoporosis	CT-RATE
15	Pericardial Effusion	CT-RATE
16	Pleural Effusion	MIMIC-CXR + CT-RATE
17	Pneumonia	MIMIC-CXR + CT-RATE
18	Pneumothorax	MIMIC-CXR
19	Possible Aspiration	MIMIC-CXR + CT-RATE
20	Possible Malignancy/Mass	CT-RATE
21	Pulmonary Artery Enlargement	CT-RATE
22	Pulmonary Edema/Vascular Congestion	MIMIC-CXR + CT-RATE
23	Pulmonary Fibrosis/Scarring	MIMIC-CXR + CT-RATE
24	Pulmonary Nodules	MIMIC-CXR + CT-RATE
25	Rib/Bone Fracture	MIMIC-CXR + CT-RATE
26	Spinal Degenerative Changes	CT-RATE

Comparison with Published Literature

To evaluate the clinical efficacy and design scope of Lumora, we compare its pipeline and model coverage against both chest X-ray diagnosis papers and recent CT-focused multimodal medical AI papers from strong publication venues. This makes the comparison fairer because Lumora supports both CXR and thoracic CT workflows.

Feature-by-Feature Comparison Matrix

Feature / Dimension	Paper 1: Alshmrani et al. (2023) [1]	Paper 2: Sharma & Guleria (2023) [2]	Paper 3: Hamamci et al. (2026) [3]	Paper 4: Zhang et al. (2025) [4]	Lumora (Ours)
Publication Quality / Venue	Alexandria Engineering Journal; Q1 engineering venue	Procedia Computer Science; indexed conference-proceedings venue	Nature Biomedical Engineering; Q1 biomedical engineering venue	Scientific Data; Q1 Nature Portfolio data journal	Academic research prototype with public model artifacts
Primary Base Model(s)	VGG19 + 3 custom CNN blocks	VGG-16 + Neural Network classifier	CT-CLIP + CT-CHAT foundation models	RadGenome-Chest CT dataset + grounded VLM baselines	5-Model Pipeline: YOLOv11s-cls + DenseNet121-GPT2 + Bio_ClinicalBERT + T5-small + LoRA
Supported Modalities	Chest X-ray (CXR) only	Chest X-ray (CXR) only	3D chest CT volumes	3D chest CT volumes with region-level grounding	Multi-Modal: Chest X-ray (CXR) AND Thoracic CT scans (2D axial slices)
Primary Task Scope	Multi-class Classification	Binary Classification	CT abnormality detection, retrieval, and CT chat	Grounded CT report generation and CT visual question answering dataset	End-to-End Clinical Flow: Guardrail → Report Gen → Label Extraction → Layperson Translation
Narrative Report Gen	No (Outputs class labels only)	No (Outputs "Pneumonia" or "Normal")	CT-CHAT supports CT-focused language interaction	Provides grounded CT report-generation supervision	Yes (Generates full, contextual English reports using visual embeddings)
Pathology Vocabulary	6 Classes (COVID-19, Normal, Pneumonia, Lung Opacity, TB, Cancer)	2 Classes (Pneumonia vs. Normal)	18 CT abnormality labels in CT-RATE plus report text	Region-grounded reports, VQA pairs, and anatomical segmentation labels	26 canonical pathologies across both X-ray and CT modalities
Patient-Friendly Translation	No	No	No dedicated layperson translation module	No dedicated layperson translation module	Yes (Rewrites complex clinical jargon using sequence-to-sequence T5 with LoRA adapters)

Feature / Dimension	Paper 1: Alshmrani et al. (2023) [1]	Paper 2: Sharma & Guleria (2023) [2]	Paper 3: Hamamci et al. (2026) [3]	Paper 4: Zhang et al. (2025) [4]	Lumora (Ours)
Safety Input Guardrails	No (Assumes correct inputs)	No (Assumes correct inputs)	Not a front-door modality guardrail	Dataset benchmark, not an input-screening pipeline	Yes (YOLOv11 guardrail checks modality and rejects non-medical images)

Key Scientific & Clinical Advancements of Lumora

- 1. Unified Clinical Workflow Integration:** Rather than acting as only a classifier or dataset benchmark, Lumora mirrors a practical radiology-support workflow. It validates the input (YOLO Guardrail), writes a narrative report (DenseNet-GPT2 VLM), identifies multiple clinical findings (ClinicalBERT), and translates these to layperson language (T5+LoRA) for direct patient use.
- 2. Cross-Modality Versatility:** The original CXR comparison papers are limited to chest radiographs, while newer CT papers focus deeply on 3D CT. Lumora sits between these lines of work by supporting both **Chest X-rays** (via MIMIC-CXR) and **Thoracic CT Scans** (via CT-RATE) in one application pipeline.
- 3. High-Granularity Disease Mapping:** Several comparator systems collapse diagnosis into a single label, binary output, or CT-specific benchmark task. Lumora classifies **26 multi-label pathologies** concurrently, capturing co-occurring conditions (e.g., Pneumonia with Pleural Effusion and Atelectasis) that are common in clinical environments.
- 4. Patient-Centric Health Literacy:** Lumora adds a patient-facing communication layer that the compared works do not include, translating jargon (e.g., "retrocardiac opacity with consolidation") into readable lay terms.

References & Citations

- [1] Alshmrani, G. M. M., Ni, Q., Jiang, R., Pervaiz, H. B., & Elshennawy, N. M. (2023). *A deep learning architecture for multi-class lung diseases classification using chest X-ray (CXR) images*. Alexandria Engineering Journal, 64, 923-935. <https://doi.org/10.1016/j.aej.2022.10.053>
- [2] Sharma, S., & Guleria, K. (2023). *A Deep Learning based model for the Detection of Pneumonia from Chest X-Ray Images using VGG-16 and Neural Networks*. Procedia Computer Science, 218, 337-346. <https://doi.org/10.1016/j.procs.2023.01.018>
- [3] Hamamci, I. E., Er, S., Wang, C., et al. (2026). *Generalist foundation models from a multimodal dataset for 3D computed tomography*. Nature Biomedical Engineering. <https://doi.org/10.1038/s41551-025-01599-y>
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